PLC task 3 – ATM700

PLC programming, communication and HMI
Program and debug in simulated PLC with virtual AGV

This compulsory PLC task will be individually examined during scheduled examination time. Each student shall execute and explain their own PLC program for the examiner.
The focus is on program quality.

1. Use the **TwinCAT Virtual AGV** project from Canvas (AGV = Automatic Guided Vehicle). Practice and get to know how it work, read enclosed **Instructions for AGV**.

2. Create an AGV control design that fulfil the following specification:
 - Move the AGV by first select the target and then press a ***Start button*** on the operator HMI.
 - Selection of target shall be done through a ***Target selection*** on the operator HMI. The ***Target selection*** may be a drop-down menu, a line for typing a letter (*A*, *B*, *C*, *D* or *H*), or any other easily understandable input. It is not accepted to have five separate buttons.
 - Changing of target in the ***Target selection*** during AGV movement shall be neglected. Also any press of the ***Start button*** during AGV movement shall be neglected.
 - A ***Moving indication*** on the operator HMI shall indicate when the AGV moves. The ***Moving indication*** may be a blinking indicator, a rotating wheel, or any other illustrative indication.
 - A ***Position display*** on the operator HMI shall present the actual position (*A*, *B*, *C*, *D* or *H*) when the AGV is in that position and stands still there. Otherwise the ***Position display*** shall be turned off. It is not accepted to have five different indicators.
 - To assist the operator a ***Ready indication*** on the operator HMI shall be turned on when it is possible to select a new target and start the AGV.
 - Whenever the AGV moves it shall be possible to immediately stop the AGV by pressing a ***Stop button*** on the operator HMI.
 - After an immediately stop it shall not be possible to give a new target or start the AGV again before a ***Reset button*** on the operator HMI has been pressed. A press on the ***Reset button*** shall move the AGV to its *HOME* position and reset the control system to run as usual again after the AGV has reached its *HOME* position.
 - A ***Stop indication*** on the operator HMI shall indicate when an immediately stop has been done until the AGV has reached its *HOME* position and the control system is ready for accepting a new target.

3. Design a proper operator HMI for controlling the AGV. The operator HMI shall contain all in and outputs described in the specification and assist the operator to run the AGV also when he/she not can see the AGV. It is not accepted to implement the operator HMI in the AGV visualisation.

4. Divide your control design in suitable functional parts and make each one as a program or function block in suitable IEC 61131-3 language. AGV communication is a typical function that will be used many times and shall therefore be implemented as a function block.
 Do not mix-up different functions in one common program or function block.

5. Execute the entire project and verify the function.
 Structure and clean up your program and function blocks to reach good programming quality and readability.

Note
The PLC programs shall be made, executed, debugged and verified in advance at school or home.
Scheduled examination time are intended for examination of the verified PLC-programs.